Yimin Gao

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EDUCATION

Ph.D., Electrical Engineering, University of Virginia — 2023 - 2027.5 (Expected)

M.S., Electrical Engineering, University of Virginia — 2021–2022

B.E., Electrical Engineering, Univ. of Cincinnati & Chongqing Univ., Dual Degree Program — 2015–2020

SUMMARY

Ph.D. student specializing in **AI/ML accelerators** and **computer architecture**, with full-stack experience across **RTL design**, **NPU/SoC system integration**, and **compilers/runtimes** on FPGA. Skilled in hardware–software co-design, mixed-precision quantization, cycle-accurate simulation, and end-to-end ML inference on custom hardware.

PROJECTS

FlexNPU — Full-Stack FPGA AI Accelerator Platform (Compiler, Runtime, RTL)

- **Full-stack system:** Built an end-to-end AI accelerator on a Xilinx Kria KV260 (Zynq SoC): custom compiler backend, custom C/C++ runtime, and a double-buffered RTL shell with AXI DMA.
- **Compiler + runtime:** Wrote custom MLIR/IREE passes lowering ML ops (GEMM, softmax, depthwise) onto accelerators, plus an ARM-host runtime and register/DMA interface that swaps RTL tiles without recompiling.
- Achieved up to **31×** end-to-end MobileBERT INT8 speedup over the on-chip ARM CPU

Precision-Scalable Bit-Serial LLM Accelerator with Mixed-Precision Quantization

- **RTL:** Designed and verified a bit-serial systolic array (3–8 bit) in Verilog supporting per-kernel precision switching in a single datapath.
- **Quantization:** Built a mixed-precision post-training quantization pipeline in Python (PyTorch) for LLM weights, matched to the hardware's supported precisions.
- **Simulation:** Built a custom **cycle-accurate simulator** with memory and energy modeling (CACTI, Ramulator) to profile end-to-end LLM inference and guide design decisions.
- Reached near-FP16 accuracy at sub-5-bit precision with up to **1.8× throughput** and **2.0× energy reduction** across LLMs (LLaMA, OPT, Phi, Yi).

Mixed-Precision AI Acceleration on a Custom RISC-V SoC

- Re-architected a 16-bit integer multiplier into a SIMD mixed-precision dot-product unit supporting INT2–INT8 and INT16 through hardware reuse and shift–add composition, integrated into a RISC-V core via ISA and microarchitecture co-design.
- Delivered **15.5–47.7×** speedup and up to **20.5 TOPS/W** on quantized CNN kernels over a baseline RISC-V.
- Integrated the AI-extended core into a full FPGA-based SoC and evaluated quantized GEMM under memory-constrained conditions, reaching up to **37×** speedup by cutting instruction count and memory traffic.

EXPERIENCE

Research Assistant, High-Performance Low-Power (HPLP) Lab, UVA — 2022–Present

- Conducted research in AI hardware, quantization, and hardware security.
- Teaching Assistant for Computer Architecture & Design, Embedded Systems, and AI Hardware courses.

SKILLS

Hardware/RTL: Verilog, SystemVerilog, Vivado, ModelSim, Synopsys DC/ICC2
Programming: C/C++, Python, RISC-V Assembly, CUDA, Tcl, Bash
ML/Compilers: PyTorch, MLIR/LLVM, IREE, model quantization, cycle-accurate simulation
Tools: Git, Make, Cadence Virtuoso (ADE, Spectre)

SELECTED PUBLICATIONS

- **Y. Gao**, et al., "FlexPosit: Precision-Scalable Bit-Serial Accelerator with Posit Quantization for Efficient LLM Inference," **MICRO 2026 (accepted)**.
- X. Zhao, **Y. Gao**, et al., **IEEE Transactions on Computers, 2024; GLSVLSI 2023**.
- **Y. Gao**, et al., LiteAIR5: a system-level framework for the design and modeling of AI-extended RISC-V cores, **IEEE SOCC 2023**.
- U.S. Patent Application: Membrane DRAM-PIM Filtering Architecture (Contributor), 2025.